Design and Development of an Intelligent Systems Framework for Multi-Agent Collaboration

A thesis submitted in partial fulfilment of the requirements

for the award of the degree of

Bachelor of Technology

By

Student Name A	**(2201CS0001)**
Student Name B	**(2201CS0002)**
Student Name C	**(2201CS0003)**
Student Name D	**(2201CS0004)**

Under the Guidance of

Dr. John Doe

Assistant Professor, Department of Computer Science & Engineering

Department of Computer Science & Engineering

Manipur Technical University

Imphal, Manipur – 795004

Month, Year

DEDICATION

This work is dedicated to our beloved parents, family members, teachers, and friends
for their constant support, guidance, and endless encouragement
throughout our academic journey.

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BONAFIDE CERTIFICATE

This is to certify that the thesis entitled **"Design and Development of an Intelligent Systems Framework for Multi-Agent Collaboration"** submitted by:

Student Name A	**(2201CS0001)**
Student Name B	**(2201CS0002)**
Student Name C	**(2201CS0003)**
Student Name D	**(2201CS0004)**

to the **Department of Computer Science & Engineering**, in partial fulfilment of the requirements for the award of the **Bachelor of Technology** degree in **Computer Science and Engineering**, is a genuine and original work carried out by them under my supervision during the academic year **202X–202X**.

The project embodies the results of their own research and development work and has not been submitted, either in part or in full, for the award of any other degree or diploma of this university or any other institution. The work has been carried out with sincerity, academic integrity, and adherence to research ethics.

We consider this work worthy of consideration for the partial fulfilment of the requirements for the said degree.

Supervisor

Dr. John Doe
Assistant Professor,
Department of Computer Science & Engineering

Head of Department

Mrs. Tayenjam Aerena
Assistant Professor & Head,
Department of Computer Science & Engineering

Manipur Technical University

(A University established under the Manipur Technical University Act, 2016)

Imphal, Manipur – 795004

DECLARATION

I hereby declare that the thesis entitled **"Design and Development of an Intelligent Systems Framework for Multi-Agent Collaboration"**, submitted to the **Department of Computer Science & Engineering, Manipur Technical University**, in partial fulfilment of the requirements for the award of the **Bachelor of Technology** degree in **Computer Science and Engineering**, is a result of my own work and effort. Under the guidance of **Dr. John Doe**, Assistant Professor, Department of Computer Science & Engineering, Manipur Technical University.

I affirm that this work has been carried out with academic honesty and integrity. All sources of information have been properly cited and acknowledged. This thesis has not been submitted earlier, either in part or in full, for the award of any degree or diploma at this or any other institution.

Date:	**Student Name A**
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First and foremost, we are highly indebted to our project supervisor, **Dr. John Doe**, Assistant Professor, Department of Computer Science & Engineering, for providing their invaluable guidance, constant supervision, constructive criticism, and support during the design and development of this project.

We express our sincere thanks to **Mrs. Tayenjam Aerena**, Assistant Professor & Head, Department of Computer Science & Engineering, for providing the necessary facilities and resources in the department to execute this work smoothly. We would also like to thank the Vice Chancellor of **Manipur Technical University** for supporting academic excellence and providing a conducive environment for research.

Lastly, we express our heartfelt gratitude to our parents, family members, classmates, and friends for their continuous encouragement and moral support throughout our project work.

Date:

Place:

Sincerely,

Student Name A

Student Name B

Student Name C

Student Name D

ABSTRACT

An abstract is a self-contained, short summary of the entire thesis. It should typically be between 200 to 300 words and present a concise overview of the problem, the methodology, the major results, and the main conclusions.

A good abstract should answer the following questions:

1. **What is the problem and why is it important?** (Context and Motivation)
2. **What did you do and how did you do it?** (Methodology / System Design)
3. **What did you find?** (Key Results and Evaluation Metrics)
4. **What are the main implications or conclusions of your work?** (Significance)

Keywords: Keyword 1, Keyword 2, Keyword 3, Keyword 4, Keyword 5.

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Chapter 1

Introduction

This chapter introduces the thesis topic, provides the motivation for the research, establishes the objectives, and outlines the organization of the rest of the document.

1.1 Background and Context

Provide a broad overview of the domain of your project. For a Computer Science project, describe the technology stack, application domain, or theoretical area. For example, if your project involves machine learning in agriculture, explain the intersection of artificial intelligence and crop classification.

A citation example: machine learning algorithms have been widely used across various domains in recent years [1], [2].

1.2 Problem Statement

State the specific problem you are trying to solve clearly and concisely. Why is this a problem? Who does it affect? What are the current limitations of existing solutions?

1.3 Objectives

Define the goals and objectives of your thesis. You can list the primary and secondary objectives using the `itemize` environment:

- To design and develop an efficient software system that solves the identified problem.

- To implement a deep learning model with high accuracy for target classification.
- To evaluate the performance of the system against baseline models.
- To deploy the application as an interactive web-based service for end-users.

1.4 Motivation

Explain why this research is worth pursuing. What are the benefits of solving this problem? Talk about the relevance to the industry, academia, or society (especially in a regional context like Manipur).

1.5 Thesis Organization

Describe how the remaining chapters of your thesis are structured. A typical layout is:

- **Chapter 2: Literature Review** presents a review of existing work and technologies related to this domain.
- **Chapter 3: System Design and Architecture** details the conceptual framework, architecture, and diagrams of the proposed system.
- **Chapter 4: Implementation Details** explains the coding, databases, API integration, and environment settings.
- **Chapter 5: Results and Discussion** showcases the experimental results, evaluations, and screenshots of the application.
- **Chapter 6: Conclusion and Future Work** summarizes the contributions of the thesis and outlines avenues for future improvements.

Chapter 2

Literature Review

This chapter presents a review of existing research, publications, and technologies that are relevant to the problem under study. It highlights the strengths, weaknesses, and gaps in current approaches, justifying the need for your proposed work.

2.1 Review of Existing Methods

Analyze previously proposed systems, frameworks, or algorithms. For instance, you can group them chronologically or by methodology (e.g., traditional machine learning vs. deep learning).

For example, author A [3] introduced Deep Residual Networks which revolutionized image classification. Meanwhile, other studies explored mobile application extensions for low-resource settings [4].

2.2 Summary of Related Work

It is highly recommended to summarize the literature review in a comparative table. Table 2.1 shows an example of how you can compare different papers based on their methodology, datasets, advantages, and limitations.

2.3 Research Gap

Based on the literature review, identify the gaps in existing systems. These gaps will define your project's novel contribution. For example:

- Most existing systems focus on theoretical models and lack real-world deploy-

Table 2.1: Comparative Summary of Related Works

Author & Year	Methodology	Advantages	Limitations
Kamilaris et al. (2018) [1]	Survey on deep learning in agriculture	Comprehensive overview of models and datasets	General review, does not propose a specific system
Mohanty et al. (2016) [2]	ResNet-50 for plant disease classification	High classification accuracy on public datasets	Poor performance on real-world backgrounds
He et al. (2016) [3]	Deep Residual Learning	Solves vanishing gradient problem	High computational and parameter cost

ability.

- There is a lack of localized datasets specific to the Northeastern region of India or Manipur.
- Existing applications lack offline capabilities or SMS-based feedback systems for rural users.

Your project aims to bridge these gaps by introducing an integrated, user-friendly, and localized system.

Chapter 3

System Design and Architecture

This chapter details the design of the proposed system. It includes the block diagram, system architecture, database schema, and mathematical formulation of the models used.

3.1 System Architecture

Describe the high-level architecture of your system (e.g., Client-Server architecture, Microservices, or MVC pattern). It is helpful to visualize the flow using block diagrams.

Figure 3.1 shows an example of how you can include a figure block in LaTeX. Make sure to place your diagrams inside the `images/` folder.

Figure 3.1: Placeholder Logo: Replace this with your System Architecture Diagram

3.2 Proposed Methodology

Detail the steps involved in your algorithm or system pipeline. You can use numbered lists or flowchart references:

1. **Data Acquisition:** Gathering raw data from sensors, users, or external APIs.
2. **Preprocessing:** Cleaning, normalising, and transforming the input data.
3. **Processing:** Applying core models, neural networks, or processing scripts.
4. **Output Generation:** Presenting the results, saving to database, or returning API responses.

3.3 Mathematical Model

Describe any mathematical formulas or loss functions used in your project. In LaTeX, you can write inline equations like $E = mc^2$ or block equations.

An example of a labeled block equation for reference:

$$\text{Softmax}(z_i) = \frac{e^{z_i}}{\sum_{j=1}^{C} e^{z_j}} \tag{3.1}$$

Equation 3.1 represents the Softmax activation function used in the final layer of our classification neural network to compute probabilities over C target classes.

Another example using the `align` environment:

$$\begin{align} \text{MSE} &= \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 \tag{3.2} \\ \text{MAE} &= \frac{1}{n} \sum_{i=1}^{n} |y_i - \hat{y}_i| \tag{3.3} \end{align}$$

These metrics are commonly used in regression models to calculate prediction errors.

Chapter 4

Implementation Details

This chapter details the actual implementation of the proposed system. It includes information about the software tools, database tables, API routes, and critical code snippets.

4.1 Technology Stack

List and describe the programming languages, libraries, frameworks, and operating systems used. For example:

Backend: Python with FastAPI or ASP.NET Core for writing high-performance REST APIs.

Frontend: TypeScript and React/Next.js for building a responsive, component-based user interface.

Database: PostgreSQL or Firebase Firestore for storing structured application data.

Machine Learning: PyTorch or TensorFlow for training and evaluating deep learning models.

4.2 Algorithms and Pseudocode

You can present the core operations of your system using pseudocode. Algorithm 1 shows an example using the `algorithm` package.

Algorithm 1 Data Processing and Verification

Require: Input dataset D, Threshold parameter τ

Ensure: Verified items list V

Initialize empty list V
for all item $x \in D$ **do**
    $score \leftarrow \text{EvaluateModel}(x)$
    if $score \geq \tau$ **then**
        Add x to V
        SendNotification(x, "Verified")
    else
        SendNotification(x, "Rejected")
    end if
end for
return V

4.3 Code Listings

You can include code listings directly in LaTeX. Listing 4.1 shows an example python function.

```python
import torch
import torch.nn as nn

def predict_sample(model, image_tensor, threshold=0.5):
    # Set model to evaluation mode
    model.eval()
    with torch.no_grad():
        output = model(image_tensor)
        probabilities = torch.sigmoid(output)
        prediction = (probabilities > threshold).float()
    return prediction.item(), probabilities.item()
```

Listing 4.1: Sample Python Inference Function

Listing 4.3 shows how to display a custom terminal/shell output using the `tcblisting` environment configured in the preamble.

Server Execution Terminal

```
$ python main.py --mode=train --epochs=10
Loading training datasets...
Training started. Epoch 1/10 - loss: 0.652 - acc: 0.723
Training completed. Model saved to outputs/best_model.
   pth
```

Chapter 5

Results and Discussion

This chapter presents the experimental findings and results of the proposed system. It compares the results with baseline models and provides a detailed discussion of the system's performance.

5.1 Experimental Setup

Describe the dataset details (size, division of training, validation, and test splits) and hardware environment (GPU, RAM) used for running tests.

5.2 Performance Evaluation

Present the performance metrics. You can tabulate classification or execution results as shown in Table 5.1.

Table 5.1: Model Performance Metrics Comparison

Model Name	**Accuracy (%)**	**Precision (%)**	**Recall (%)**	**F1-Score (%)**
Baseline CNN	82.4	80.1	81.3	80.7
VGG16 Transfer	89.2	88.5	89.0	88.7
ResNet-50 (Proposed)	**95.6**	**94.8**	**95.2**	**95.0**

5.3 Analysis of Visual Results

You can present training curves or other visual outputs side-by-side using the `subfloat` command. Figure 5.1 shows an example placeholder.

(a) Accuracy Curve

(b) Loss Curve

Figure 5.1: Training and Validation Performance Curves (Placeholders)

5.4 Discussion

Explain what the results mean. Why did your model perform better? Are there any scenarios where the system failed or showed low confidence? Discuss the practical implications of these results for the end-user.

Chapter 6

Conclusion and Future Work

This chapter summarizes the overall contributions of this thesis and outlines promising directions for future research.

6.1 Summary of Work Done

Review the problem statement and highlight how the objectives defined in Chapter 1 have been met. Summarize the design, development, and testing phases.

Key achievements include:

- Successful implementation of a robust software architecture for the target problem.
- Development of high-accuracy computational models.
- Deployment of the system as an accessible, real-time application.

6.2 Contributions of the Project

State the unique value your project brings. This could be a new dataset, an open-source library, a practical tool for society, or an optimized algorithm.

6.3 Limitations of the Current System

Be honest about what the current system cannot do. For example:

- The system requires a continuous internet connection, limiting usage in remote areas.

- The classification model is trained on a limited set of classes and may not generalize well to unseen categories.
- Processing latency increases under high concurrent user load.

6.4 Future Work

Provide clear, actionable directions for future students or researchers who want to build upon your work.

- Extending the dataset to cover a wider variety of classes.
- Optimizing the model size for edge devices (such as mobile phones) to allow offline deployment.
- Integrating security improvements, such as blockchain-based auditing.

REFERENCES

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[2] S. P. Mohanty, D. P. Hughes, and M. Salathé, "Using deep learning for image-based plant disease detection," *Frontiers in Plant Science*, vol. 7, p. 1419, 2016. DOI: `10.3389/fpls.2016.01419`.

[3] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, IEEE, 2016, pp. 770–778. DOI: `10.1109/CVPR.2016.90`.

[4] P. Muñoz, M. Navas, *et al.*, "ICT and mobile applications for agricultural extension in low-resource settings: A systematic review," *Computers and Electronics in Agriculture*, vol. 154, pp. 234–245, 2018.

Chapter A

Source Code Directory Structure

This appendix documents the directory structure of the proposed system's monorepo, outlining the organization of the frontend application, backend services, and machine learning components.

Project Folder Layout

The complete system is organized into a modular monorepo structure. Below is the simplified tree representation of the codebase:

System Directory Structure

```
my-project-monorepo/
|-- backend/
|   |-- src/
|   |   |-- controllers/       # Handles incoming HTTP API request
|   |   |-- models/            # Database entities and data schema
|   |   |-- services/          # Core business logic processing
|   |   |-- config/            # Application settings and environm
|   |-- Dockerfile             # Multi-stage container specificati
|-- frontend/
|   |-- public/                # Static assets (images, icons)
|   |-- src/
|   |   |-- components/        # Reusable UI components
|   |   |-- hooks/             # Custom React/Vue hooks
|   |   |-- state/             # Centralized state management
|   |   |-- App.js             # Application entry component
|   |-- package.json           # Frontend dependencies list
|-- ml_service/
|   |-- models/                # Saved weights and model checkpoin
|   |-- predict.py             # Inference pipeline code
|   |-- train.py               # Model training script
|   |-- requirements.txt       # Python environment dependencies
|-- docker-compose.yml         # Dev and production orchestrator f
|-- README.md                  # Quick-start documentation guide
```

Chapter B

REST API Specification

This appendix provides the REST API contract specifications for the system's core services. The architecture decouples the frontend client from the backend and machine learning services using standardized HTTP request and response structures.

1. User Authentication Endpoint

POST /api/v1/auth/login

Authenticates users and returns a JSON Web Token (JWT) for secure session management.

Table B.1: Specification for `POST /api/v1/auth/login`

Parameter	Type / Format	Description
Headers	`application/json`	Request payload must be formatted as JSON.
Payload: username	String (Required)	Username or email address of the registered user.
Payload: password	String (Required)	Case-sensitive raw password.
Success Response	`200 OK`	Returns: `{ "token": "JWT_STRING", "expires": 3600 }`
Error Response	`401 Unauthorized`	Triggered by invalid credentials or inactive accounts.

2. Machine Learning Inference Endpoint

POST /api/v1/predict

Executes real-time model inference on user-uploaded imagery or sensor telemetry.

Table B.2: Specification for `POST /api/v1/predict`

Parameter	**Type / Format**	**Description**
Content-Type	`multipart/form-data`	Requires binary file upload stream.
Payload: file	Binary File (Required)	Image or data file (e.g., JPEG, PNG, CSV).
Payload: threshold	Float (Optional)	Confidence threshold. Defaults to `0.50`.
Success Response	`200 OK`	Returns JSON listing prediction classes, confidence scores, and processing latency.
Error Response	`422 Unprocessable`	Raised if the input file format is unsupported or corrupted.

Chapter C

Database Schema Dictionary

This appendix documents the database schema and data dictionary supporting the application's persistence layer. Tables are designed in Third Normal Form (3NF) to ensure integrity.

1. Users Table (`users`)

Stores core user details, credential hashes, and system access role designations.

Table C.1: Data Dictionary for `users` Table

Column Name	**Data Type**	**Nullable**	**Description & Constraints**
`id`	`UUID`	No	Unique identifier (Primary Key).
`username`	`VARCHAR(50)`	No	Unique username for authentication.
`email`	`VARCHAR(100)`	No	Unique email address.
`password_hash`	`VARCHAR(255)`	No	Secure Argon2 or bcrypt salted hash value.
`role`	`VARCHAR(20)`	No	System authorization role (e.g., 'Admin', 'User').
`created_at`	`TIMESTAMP`	No	Datetime of record creation. Defaults to current time.

2. Transactions Log Table (`activity_logs`)

Maintains a system audit trail to monitor requests and resource access details.

Table C.2: Data Dictionary for activity_logs Table

Column Name	Data Type	Nullable	Description & Constraints
log_id	BIGINT	No	Auto-incrementing identifier (Primary Key).
user_id	UUID	Yes	Reference key (Foreign Key linking to users.id).
action	VARCHAR(100)	No	Description of action executed (e.g., 'FILE_UPLOAD').
ip_address	VARCHAR(45)	No	Origin IP address of client requesting request.
timestamp	TIMESTAMP	No	Datetime when log was written.

Chapter D

System Configuration Parameters

This appendix details the deep learning model hyperparameters, neural network configuration parameters, and environmental baselines compiled during system optimization.

1. Model Hyperparameter Configurations

The neural network backbone parameters and training schedules are documented below:

Table D.1: Neural Network Architecture and Hyperparameter Optimization Settings

Hyperparameter	Configured Value	Engineering Rationale / Execution Notes
Backbone Network	`ResNet-50 / ViT`	Standard deep architecture options for feature extraction.
Input Resolution	$224 \times 224 \times 3$	Normalized RGB dimensions matching standard dataset structures.
Batch Size	32	Chosen to balance memory limitations and gradient stability.
Optimizer	`AdamW`	Incorporates decoupled weight decay to minimize overfitting.
Initial Learning Rate	1×10^{-4}	Paired with a cosine annealing learning rate scheduler.
Dropout Rate	0.3	Introduced in fully connected layers to prevent co-adaptation.

2. Environmental Baselines

The application services depend on the configuration baselines listed below:

Table D.2: System Environment Variables and Server Configs

Variable Key	**Default Value**	**Usage & Constraint Details**
ENV	production	Configures security policies and levels of logging detail.
PORT	8080	Server port bound by application daemon.
DB_MAX_POOL	20	Maximum connection pool allotment allowed for DB driver.
TIMEOUT_SEC	30	Active threshold timing out slow network sockets.

Chapter E

Deployment and Containerization Specifications

This appendix details the containerization scripts and system orchestrator file structures required to build, test, and deploy the application ecosystem across containerized cloud runtimes.

1. Dockerfile Specification

Below is the Dockerfile template utilized to containerize the Python-based API service:

```
# Use lightweight base Python image
FROM python:3.11-slim as builder

WORKDIR /app

# Install system dependencies
RUN apt-get update && apt-get install -y \
    build-essential \
    && rm -rf /var/lib/apt/lists/*

# Install python dependencies in a virtual env
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

# Final production stage
FROM python:3.11-slim as runtime
WORKDIR /app
COPY --from=builder /usr/local/lib/python3.11/site-packages /usr/
    local/lib/python3.11/site-packages
```

```
COPY --from=builder /usr/local/bin /usr/local/bin
COPY . .

# Expose web server port
EXPOSE 8080

CMD ["uvicorn", "main:app", "--host", "0.0.0.0", "--port", "8080"]
```

2. Orchestration Specification (Docker Compose)

Below is the configuration template for orchestrating multi-container systems locally:

```
version: '3.8'

services:
  web:
    build: ./backend
    ports:
      - "8080:8080"
    environment:
      - DATABASE_URL=postgresql://user:pass@db:5432/mydb
    depends_on:
      - db

  db:
    image: postgres:15-alpine
    volumes:
      - pgdata:/var/lib/postgresql/data
    environment:
      - POSTGRES_USER=user
      - POSTGRES_PASSWORD=pass
      - POSTGRES_DB=mydb
    ports:
      - "5432:5432"

volumes:
  pgdata:
```